



HiLCoE

**School of Computer Science and
Technology**

**Patient Record and Appointment Management
System for Addis Ababa Private Clinics**

SOFTWARE REQUIREMENTS ANALYSIS AND SPECIFICATION

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Definitions, Acronyms, and Abbreviations

Term	Definition
PRAAMS	Patient Record and Appointment Management System
SRS	Software/System Requirements Specification
HiLCoE	Institution offering this Senior Project
UI	User Interface.
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
RBAC	Role-Based Access Control
RLS	Row-Level Security
UUID	Universally Unique Identifier
API	Application Programming Interface.
TLS	Transport Layer Security
CRUD	Create, Read, Update, Delete
UAT	User Acceptance Testing

1. Introduction / Overview

This is the Software/System Requirements Analysis and Specification Document for the Patient Record and Appointment Management System for Addis Ababa Private Clinics ("PRAAMS"). It is based on the Senior Project Proposal Document previously prepared and approved, and includes the results of the requirements elicitation and analysis activities performed since then, including reviewing the current manual process used by private clinics and interacting with them (via interviews).

This document thus specifies the scope of the proposed system and solution in terms of requirements to be met by the system (functional and non-functional) and analysis-level descriptions of system features (OOSE scenarios, use case diagram, use case descriptions, an object model, and a dynamic model) in accordance with the requirements of the School's recommended Object-Oriented Software Engineering approach. In addition, it includes low-fidelity mock-ups of the proposed user interface and navigation paths between different screens. All of these are used to define the system boundary and system functionality to be designed and implemented.

2. Current System

At present, most of the private clinics in Addis Ababa are using a manual system to manage information about their patients and appointments. Patient's information records are stored manually and appointments are made on phone or face to face and recorded in a notebook. There are several shortcomings in the current private clinic management system which motivate the need for developing a new system. Those include:

Searching patient's information that is stored in paper takes a lot of time especially if the number of patients is large

The patient's information stored manually may be lost or torn or there might be duplication of information in different manual files. Hence, for some patients, it might be necessary to resubmit their records.

Appointments might be scheduled twice for the same time or forgotten because there is no central place or person to coordinate the scheduling of the appointments. This causes wastage of time for the patients who come to the clinic but cannot be attended to immediately

Records are not neat

It is not possible to prevent unauthorized access to patient's information or to know who has accessed a particular record.

It is not possible to make statistical reports about the performance of each clinic based on their records conveniently.

The shortcomings of the current system motivated us to come up with the requirements specified in the subsequent sections.

3. Proposed System

The proposed Patient Record and Appointment Management System captures, stores, and manages the registration, medical records, and appointment schedules of private clinics in Addis Ababa as a centralized role-based web application automating their current manual process described in the above section.

3.1. Function Definition

At a high level, the functions of the proposed system fall into such categories as those described below:

Patient registration : registering new patients and capturing their personal and contact details in their records.

Electronic patient record management : managing electronic patient records, including creating/updating and storing patients' medical records in a secure database.

Appointment management : scheduling, rescheduling or cancelling appointments with healthcare professionals based on their availability and recording all scheduled appointments in the system.

Patient record lookup: searching for and retrieving patient records.

Report generation : generating reports related to the registration, appointment, and staff activity.

User authentication and role-based access control : verifying the identity of users who enter the authentication screen and ensuring that they are only provided with the functionality and data they are authorized to use.

Each of these functions is further described under Activities in the Use Case Descriptions section (3.4.3) against the input/process/output, precondition/postcondition.

3.2. Functional Requirements

Table 1 lists the functional requirements of the system, grouped by subsystem. Each is phrased as "the system shall provide/do," in line with the documentation guideline.

Table 1: Functional Requirements

ID	Requirement
User Management & Access Control	
FR-01	The system shall allow an Administrator to create, update, deactivate, and delete staff user accounts (Receptionist, Healthcare Professional, Administrator) roles.
FR-02	The system shall authenticate users by email and password to log in the system.
FR-03	The system shall allow a user to access only those functions and information available to their user role.
FR-04	The system shall allow an Administrator to configure and manage user account role permissions.
FR-05	The system shall automatically terminate a user session after a defined period inactivity.
Patient Record Management	
FR-06	The system shall allow a Receptionist to register a new patient by adding demographic and contact information.
FR-07	The system shall check for potential duplicate patient records during registration.
FR-08	The system shall allow authorized users to update existing patient information.
FR-09	The system shall allow authorized users to search for and retrieve patient records by name, patient ID, or phone number.
FR-10	The system shall allow a Healthcare Professional to add medical record entries to a patient's file.
FR-11	The system shall allow a Healthcare Professional to view a patient's complete medical history.
FR-12	The system shall maintain a permanent, non-editable audit log of changes made to a patient's medical record.
Appointment Scheduling & Reporting	

FR-13	The system shall allow a Receptionist to schedule an appointment for a patient with an available healthcare professional.
FR-14	The system shall display available time slots for each healthcare professional.
FR-15	The system shall prevent double-booking of the same time slot.
FR-16	The system shall allow a Receptionist to reschedule or cancel an existing appointment.
FR-17	The system shall allow Receptionists and Healthcare Professionals to view a calendar of scheduled appointments.
FR-18	The system shall allow an Administrator to generate reports on patient registrations, appointment statistics, and staff activity within a specified date range.
FR-19	The system shall allow generated reports to be viewed on-screen, printed, or exported as PDF or CSV.

3.3. Non-Functional Requirements

Table 2 lists the non-functional requirements that constrain how the system must perform its functions. They will be described in this system's context.

Table 2: Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall retrieve and display a patient record within 3 seconds under normal network conditions.
NFR-02	Security	All sensitive patient data shall be encrypted in transit and at rest.
NFR-03	Security	User passwords shall be stored using a strong one-way hashing algorithm and plaintext passwords shall never be stored or logged.
NFR-04	Availability	The system shall be operational at least 99% of the time.

NFR-05	Usability	The interface shall be simple enough that a user with basic computer literacy can register a patient after a short orientation, without formal technical training.
NFR-06	Reliability	The system shall not lose data already committed to the database in the event of a browser crash or temporary network interruption.
NFR-07	Maintainability	The system shall be built using modular components to simplify future maintenance debugging, and feature additions.
NFR-08	Scalability	The database design shall support continued growth in patient and appointment records without significant lose in performance.
NFR-09	Compliance	The system shall comply with applicable Ethiopian data protection requirements.
NFR-10	Auditability	The system shall record an audit of key actions, like login attempts, record edits, and access to patient files that help accountability.

3.4. The System Models

This section describes the requirements and analysis models of the new system using the OOSE approach: scenarios, a use case diagram, use case descriptions, a activity diagrams, and a class diagram.

3.4.1. Scenarios

The following scenarios show typical end-to-end interactions

Table 3: Scenario 001

Scenario	Description
Patient Registration and Appointment Booking	A new patient walks into a private clinic in Addis Ababa seeking a consultation. The receptionist searches the system and confirms no existing record for the patient. The receptionist registers the patient by entering their demographic and contact details, then the system generates a unique patient ID. The receptionist then selects an available healthcare professional and slot, enters the reason and confirms the appointment. The system creates the

	appointment record and displays a confirmation, which the receptionist relays to the patient.
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Table 4: Scenario 002

Scenario	Description
Scenario 002 - Consultation and Medical Record Update	A healthcare professional accesses the database by searching for the patient's name or opening their record. The patient's personal medical history is opened for perusal, with all previous diagnoses available for review. After the consultation, the healthcare professional creates a new medical record entry, including the diagnosis, notes, and the date of the appointment. The new record is saved and permanently added to the patient's medical history, with the date and professional who made the entry noted for accountability.

Table 5: Scenario 003

Scenario	Description
Scenario 003 - Administrative Reporting and Staff Account Management	At the end of the month, the clinic administrator accesses his account to check the organization's performance. He prints the report with all registered patients and their appointments for the month and provides it to the management team for examination. Meanwhile, the administrator creates a new user account for the newly hired receptionist, giving him/her the necessary roles to access the appropriate functions: registering patients and scheduling appointments.

Table 6: Scenario 004

Scenario	Description
Scenario 004 - Unauthorized Access Attempt	A receptionist, being logged on to the system, decides to check out the administrative reporting module for fun. Because the receptionist does not have access rights to administrative reports, the system will reject the request and notify the receptionist that he or she does not have permission to view the information. The receptionist's inquiry will be documented in the system logs for review by the administrator.system's audit log, which the administrator can later review as part of routine security monitoring.

3.4.2. Use Case Diagram

The system's use cases are organized into three subsystems, each modeled as a separate use case diagram for clarity: User Management & Access Control, Patient Record Management, and Appointment Scheduling & Reporting.

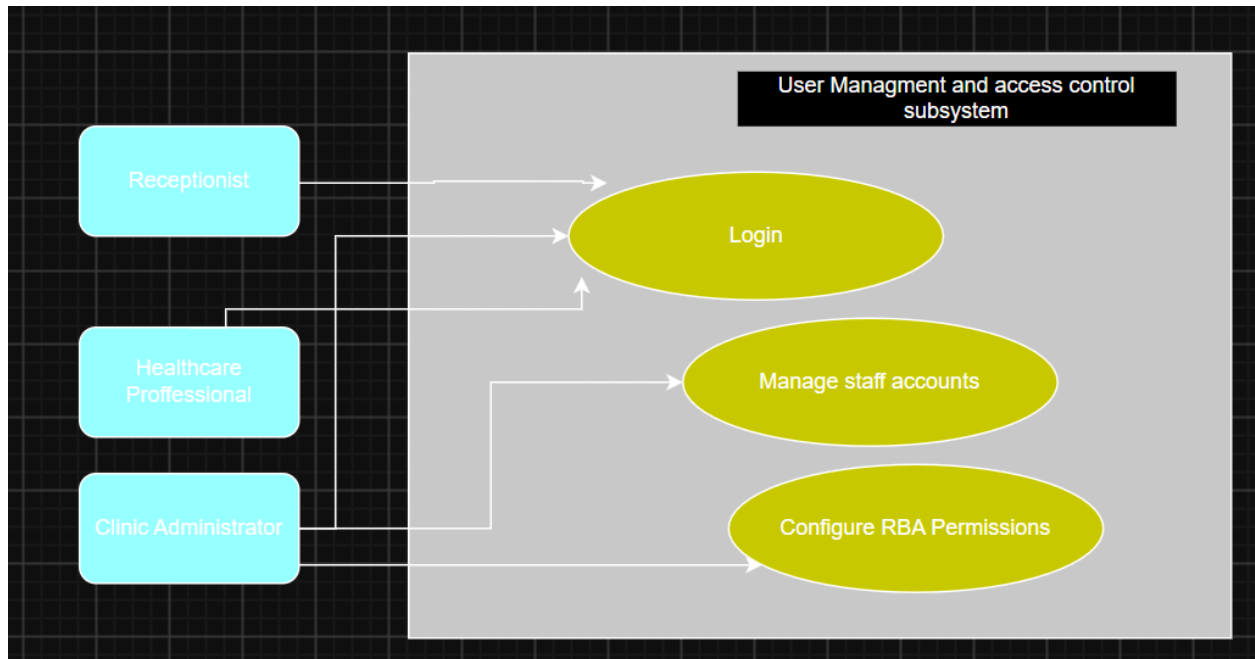


Figure 1: User Management & Access Control Subsystem – Use Case Diagram

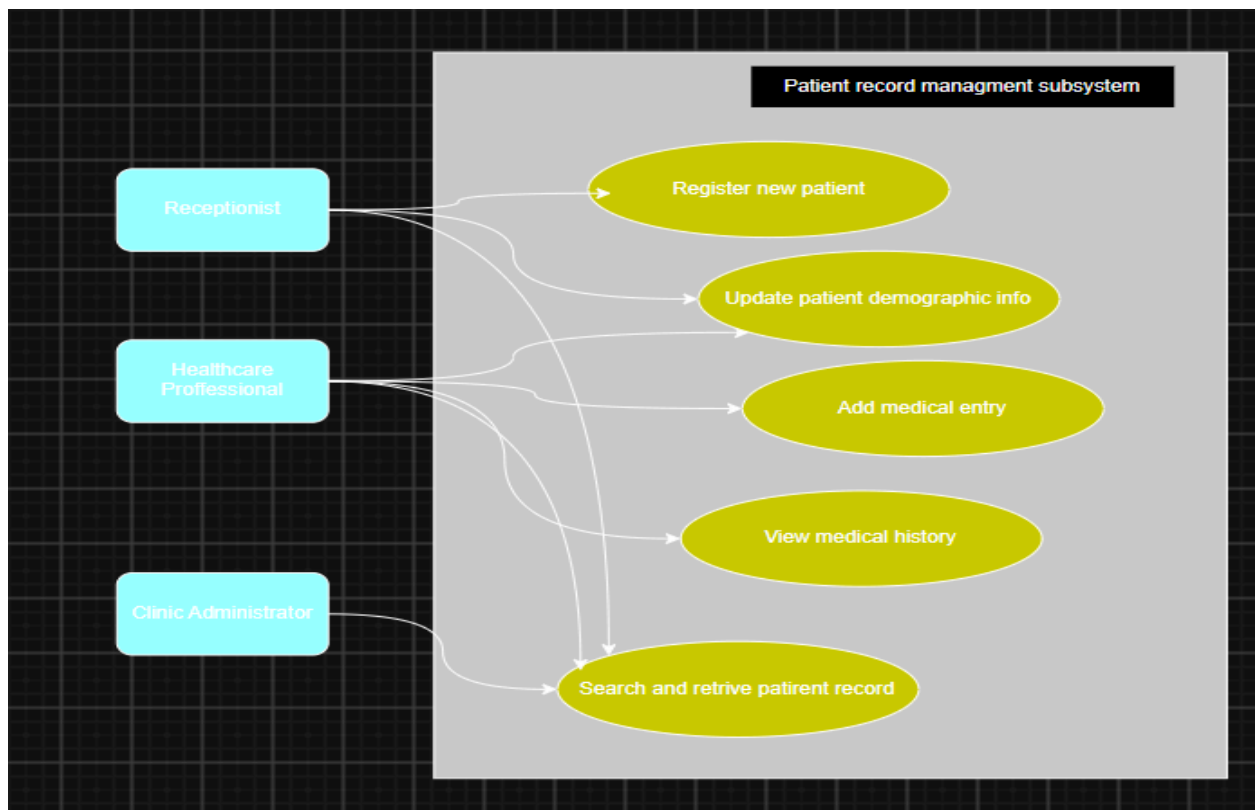


Figure 2: Patient Record Management Subsystem – Use Case Diagram

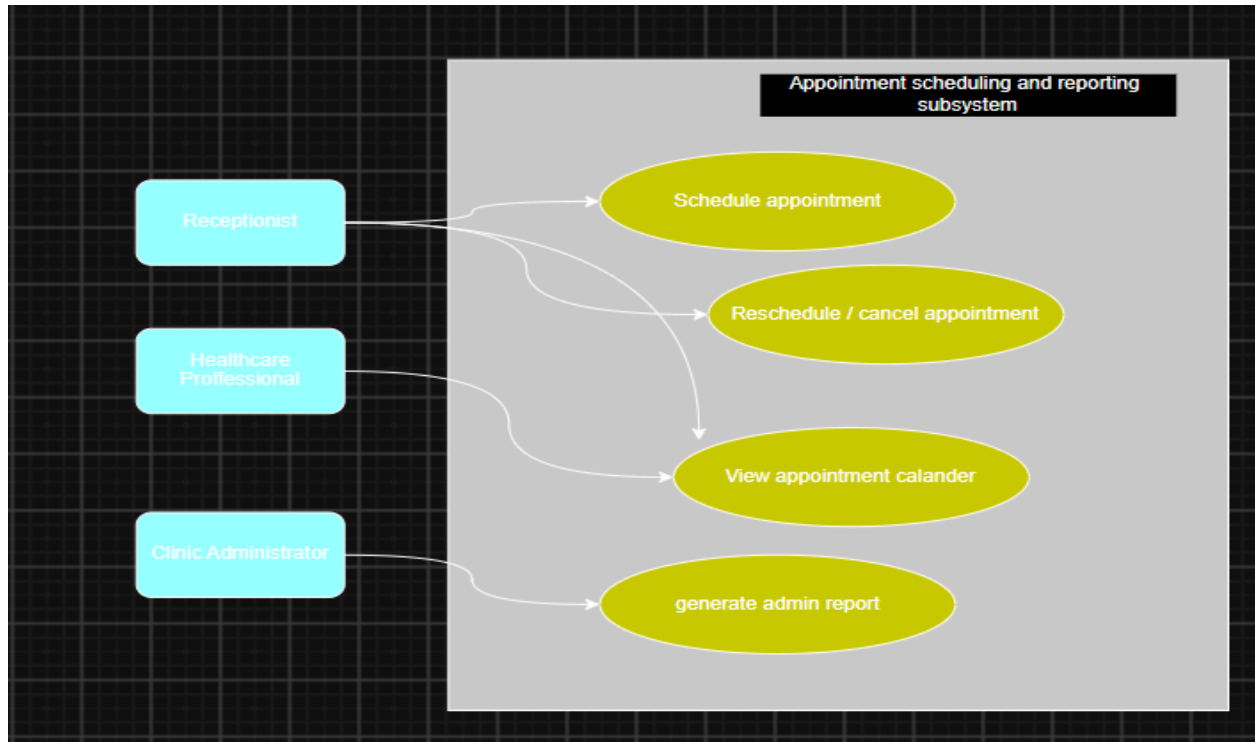


Figure 3: Appointment Scheduling & Reporting Subsystem – Use Case Diagram

3.4.3. Use Case Descriptions

Tables 7 through 18 provide detailed descriptions of each use case identified in the diagrams above including actors, preconditions, basic and alternative courses of action, and postconditions.

Table 7: Use Case UC-01 – Login

Field	Detail
Actor(s)	Receptionist, Healthcare Professional, Administrator
Description	Allows a registered staff member to authenticate into the system using their email and password.
Precondition	The user has a valid, active account in the system.
Basic Course of Action	1. User navigates to the login page. 2. User enters their registered email and password. 3. User submits the login form. 4. System validates the credentials with stored account records.

	5. System identifies the user's role and redirects them to their dashboard.
Alternative Course of Action	4a. If credentials are invalid, the system displays an error message and prompts the user to retry. 4b. After 5 consecutive failed attempts, the system temporarily locks the account for security.
Postcondition	The user is authenticated and redirected to their role-specific dashboard.

Table 8: Use Case UC-02 – Manage Staff Accounts

Field	Detail
Actor(s)	Administrator
Description	Allows the administrator to create, update, or deactivate staff user accounts.
Precondition	Administrator is logged in.
Basic Course of Action	1. Administrator navigates to the User Management module. 2. Administrator selects "Add New Staff" or selects an existing account to edit. 3. Administrator enters or updates staff details (name, email, role, phone). 4. Administrator submits the form. 5. System validates and saves the account information.
Alternative Course of Action	4a. If the email address is already registered, the system displays a duplicate-account error and prevents submission.
Postcondition	The staff account is created, updated, or deactivated in the system.

Table 9: Use Case UC-03 – Configure Role-Based Access Permissions

Field	Detail
Actor(s)	Administrator
Description	Allows the administrator to define which system functions each role is permitted to access.
Precondition	Administrator is logged in.

Basic Course of Action	1. Administrator navigates to Access Control Settings. 2. Administrator selects a role (Receptionist, Healthcare Professional). 3. Administrator enables or disables specific permissions for that role. 4. Administrator saves the changes. 5. System applies the updated permission set to all users assigned that role.
Alternative Course of Action	N/A
Postcondition	Role permissions are updated and enforced system-wide.

Table 10: Use Case UC-04 – Register New Patient

Field	Detail
Actor(s)	Receptionist
Description	Allows the receptionist to create a new patient record in the system.
Precondition	Receptionist is logged in.
Basic Course of Action	1. Receptionist selects "Register New Patient". 2. Receptionist enters the patient's demographic and contact information. 3. Receptionist submits the form. 4. System checks for potential duplicate records. 5. System creates the patient record and generates a unique patient ID. 6. System displays a confirmation with the new patient ID.
Alternative Course of Action	3a. If required fields are missing or invalid, the system highlights the errors and prevents submission. 4a. If a possible duplicate is found, the system flags it for receptionist review before creating a new record.
Postcondition	A new patient record exists in the system.

Table 11: Use Case UC-05 – Update Patient Demographic Information

Field	Detail
Actor(s)	Receptionist, Healthcare Professional

Description	Allows authorized staff to update an existing patient's demographic or contact details.
Precondition	The patient record exists; the user is logged in.
Basic Course of Action	1. User searches for and opens the patient record. 2. User selects "Edit Information". 3. User updates the relevant fields. 4. User saves the changes. 5. System validates and updates the record.
Alternative Course of Action	3a. If updated fields are invalid, the system highlights the errors and prevents saving.
Postcondition	The patient's information is updated in the database.

Table 12: Use Case UC-06 – Search / Retrieve Patient Record

Field	Detail
Actor(s)	Receptionist, Healthcare Professional, Administrator
Description	Allows an authorized user to locate and view a patient's record.
Precondition	User is logged in.
Basic Course of Action	1. User enters a search term (name, patient ID, or phone number). 2. System queries the database and returns matching results. 3. User selects the correct patient from the results list. 4. System displays the patient's full record.
Alternative Course of Action	2a. If no matches are found, the system displays a "no results found" message.
Postcondition	The requested patient record is displayed to the user.

Table 13: Use Case UC-07 – Add Medical Record Entry

Field	Detail
Actor(s)	Healthcare Professional
Description	Allows a healthcare professional to add a diagnosis and notes entry to a patient's medical history following a consultation.
Precondition	The patient record exists; the Healthcare Professional is logged in.

Basic Course of Action	1. Healthcare Professional opens the patient's record. 2. Healthcare Professional selects "Add Medical Entry". 3. Healthcare Professional enters the visit date, diagnosis, and clinical notes. 4. Healthcare Professional submits the entry. 5. System saves the entry, timestamps it, and links it to the professional's identity.
Alternative Course of Action	N/A
Postcondition	A new, permanent medical record entry is attached to the patient's history.

Table 14: Use Case UC-08 – View Medical History

Field	Detail
Actor(s)	Healthcare Professional
Description	Allows a healthcare professional to review a patient's full medical history in chronological order.
Precondition	The patient record exists.
Basic Course of Action	1. Healthcare Professional opens the patient's record. 2. Healthcare Professional selects "Medical History". 3. System retrieves and displays all past entries, most recent first.
Alternative Course of Action	N/A
Postcondition	The patient's medical history is displayed.

Table 15: Use Case UC-09 – Schedule Appointment

Field	Detail
Actor(s)	Receptionist
Description	Allows the receptionist to book an appointment for a patient with a healthcare professional.
Precondition	The patient record exists; the Receptionist is logged in.
Basic Course of Action	1. Receptionist searches for and selects the patient.

	2. Receptionist selects a healthcare professional and preferred date. 3. System displays available time slots. 4. Receptionist selects a slot and enters the reason for the visit. 5. System verifies the slot is still available. 6. System creates the appointment with status "Scheduled".
Alternative Course of Action	5a. If the slot was taken in the meantime, the system notifies the receptionist and refreshes the available slots.
Postcondition	The appointment is created and appears on the appointment calendar.

Table 16: Use Case UC-10 – Reschedule / Cancel Appointment

Field	Detail
Actor(s)	Receptionist
Description	Allows the receptionist to modify or cancel an existing appointment.
Precondition	An appointment exists.
Basic Course of Action	1. Receptionist locates the appointment via search or calendar view. 2. Receptionist selects "Reschedule" or "Cancel". 3. For reschedule, the receptionist selects a new available slot; the system updates the appointment. 4. For cancel, the receptionist confirms cancellation; the system updates the status to "Cancelled".
Alternative Course of Action	N/A
Postcondition	The appointment's date, time, or status is updated accordingly.

Table 17: Use Case UC-11 – View Appointment Calendar

Field	Detail
Actor(s)	Receptionist, Healthcare Professional
Description	Allows staff to view scheduled appointments in a calendar or list view.

Precondition	User is logged in.
Basic Course of Action	1. User navigates to the Appointments module. 2. User selects a date range or view (day, week, or month). 3. System displays all matching appointments with patient and status details.
Alternative Course of Action	N/A
Postcondition	The requested appointment schedule is displayed.

Table 18: Use Case UC-12 – Generate Administrative Report

Field	Detail
Actor(s)	Administrator
Description	Allows the administrator to generate summary reports on clinic activity.
Precondition	Administrator is logged in.
Basic Course of Action	1. Administrator navigates to the Reports module. 2. Administrator selects a report type and date range. 3. Administrator requests report generation. 4. System aggregates the relevant data and builds the report. 5. System displays the report; the administrator may view, print, or export it.
Alternative Course of Action	4a. If no data is found for the selected range, the system displays a "no records found" message.
Postcondition	The report is generated and made available to the administrator.

3.4.4. Activity Diagrams

Activity diagrams for four representative use cases

Figure 4: Activity Diagram for 'Login' Use Case

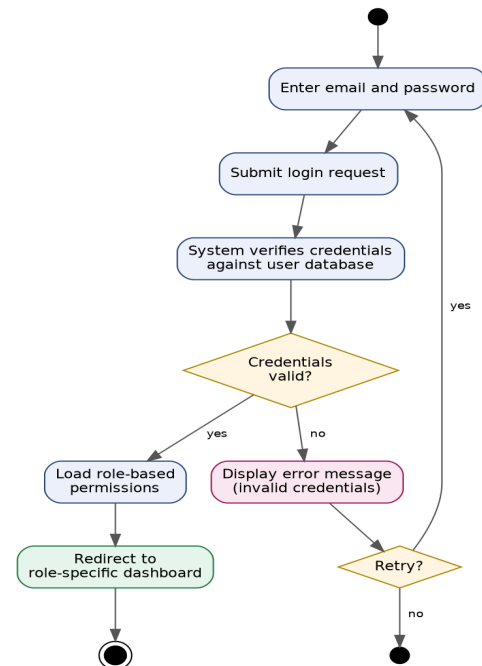


Figure 5: Activity Diagram for 'Register New Patient' Use Case

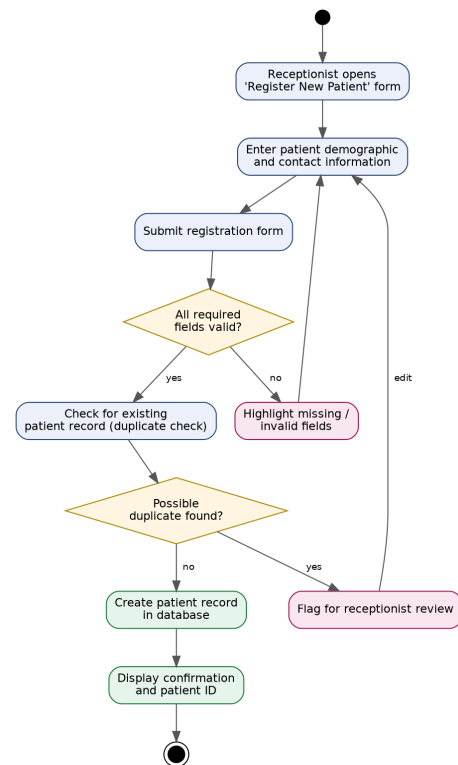


Figure 6: Activity Diagram for 'Schedule Appointment' Use Case

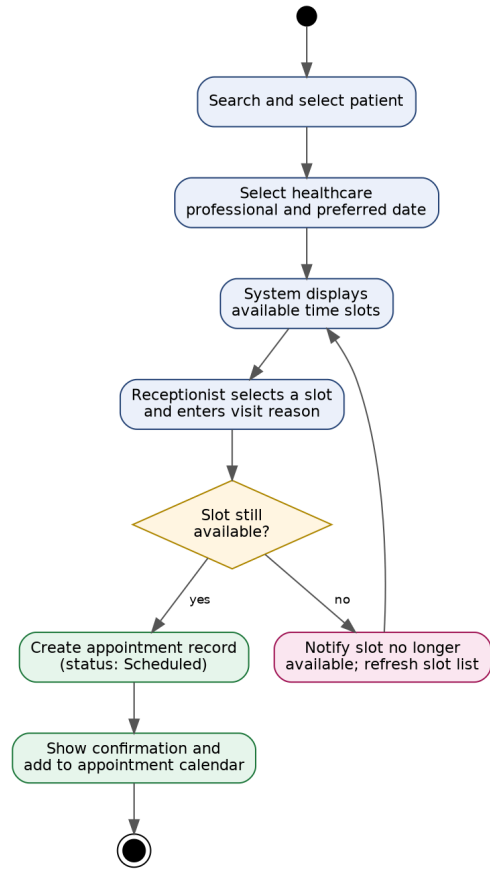
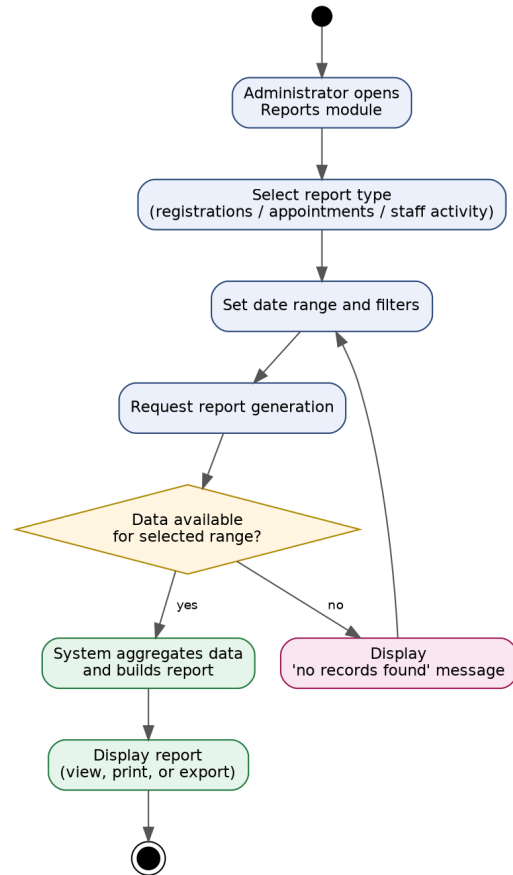


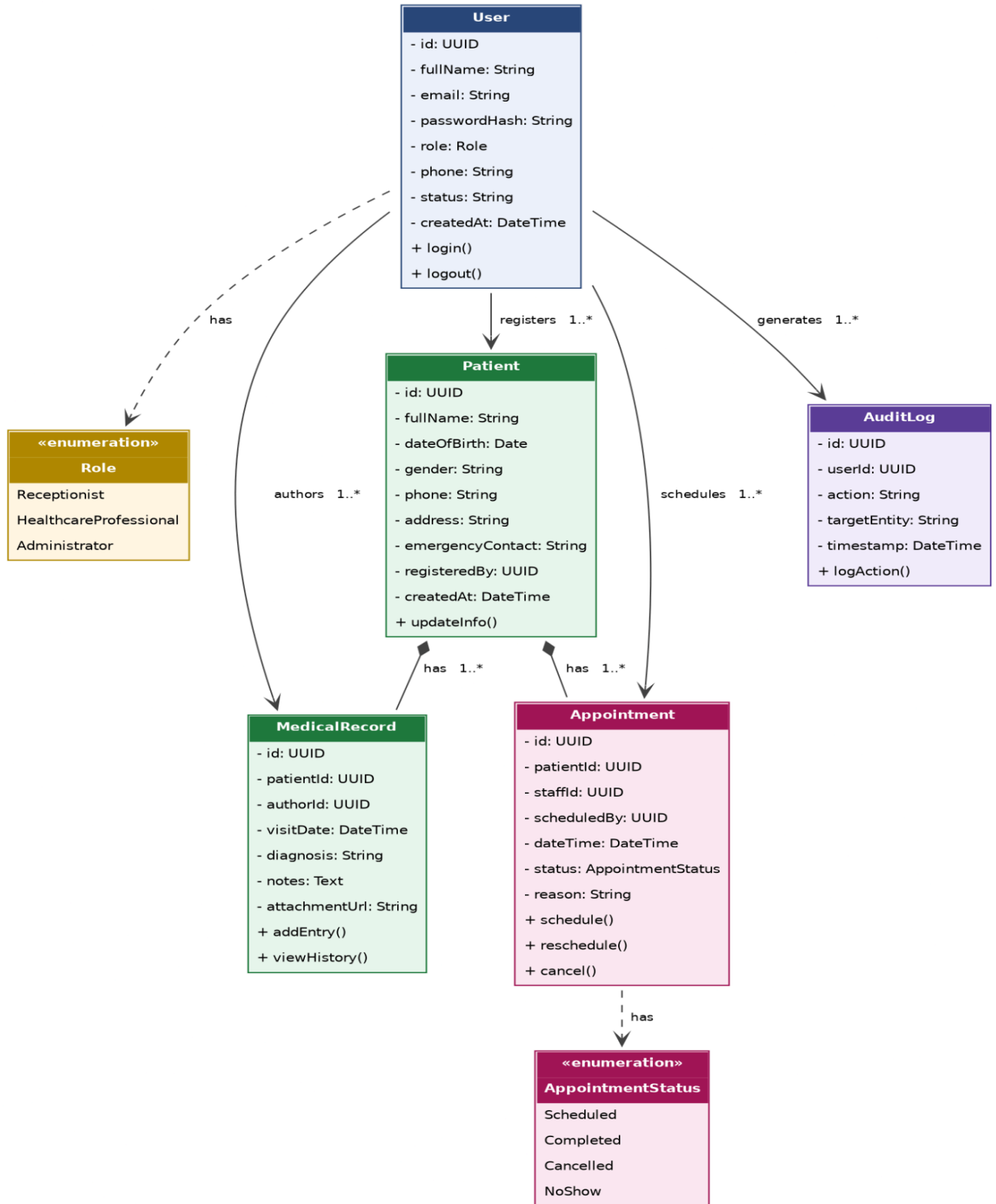
Figure 7: Activity Diagram for 'Generate Administrative Report' Use Case



3.4.5. Class Diagram

The class diagram below represents the system's main entities. The User class with the Role attribute stores information about all system staff since all of them have the same set of permissions, and the difference between an Administrator, Receptionist, and Healthcare Professional is managed by assigning different roles to users. The Patient, MedicalRecord, and Appointment classes store the main data of the application, and the AuditLog class is used to meet the requirements of reference number NFR-10.

Figure 8: Class Diagram for the Patient Record and Appointment Management System

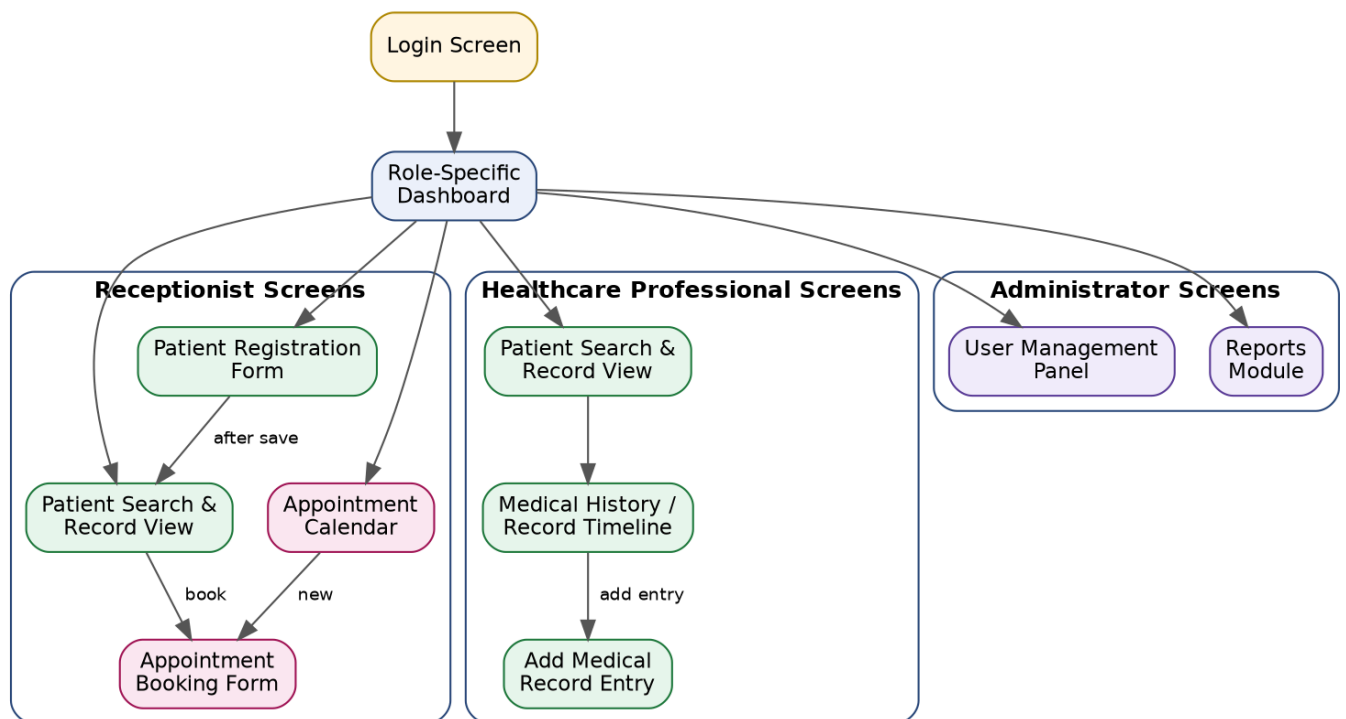


4. User Interface

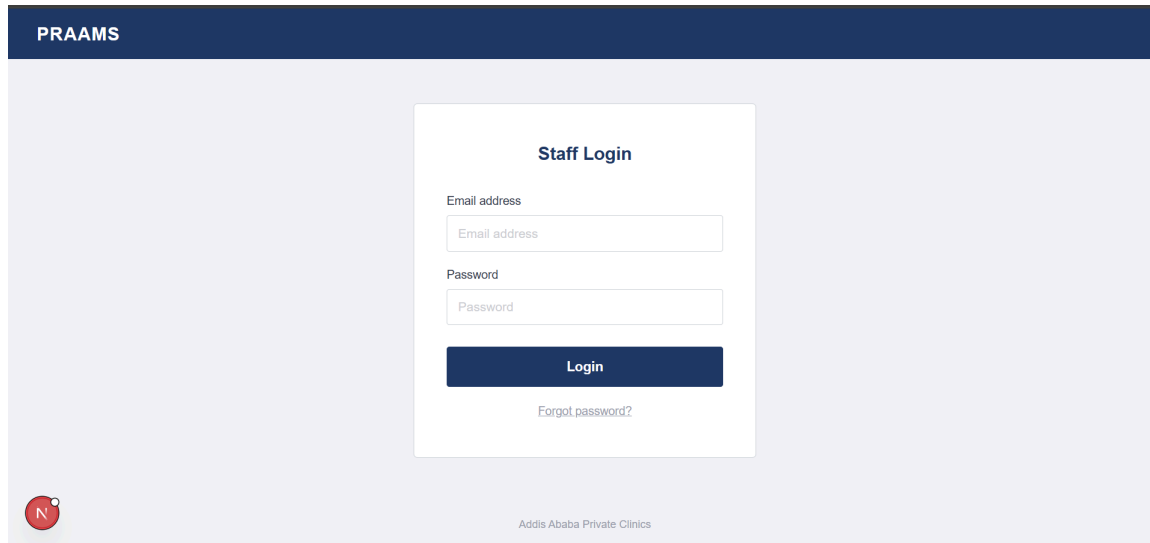
The user interface will be implemented as a responsive web application (Next.js, TypeScript, Tailwind CSS) which will be used by desktop and tablet browsers at the clinic reception and consultation desks. Below are the navigation flow between the application's screens, low-fidelity mock-ups of the key screens identified, and an inventory of all the intended screens. The following are only preliminary designs; high-fidelity mock-ups, and subsequently screenshots of the actually implemented screens, will be provided during the Design and Implementation stages, respectively.

Navigational Flow

Figure 9 below represents the navigational flow. It shows the order in which users access particular functions through a series of screens starting with the log-in and branching out according to the selected role..



Low-Fidelity Mock-ups



The mock-up shows a web application for 'PRAAMS' with a dark blue header. The main content area is light gray. A white card in the center is titled 'Staff Login'. It contains two input fields: 'Email address' and 'Password'. Below these is a dark blue 'Login' button. A link 'Forgot password?' is positioned below the button. In the bottom left corner, there is a red circular logo with a white 'N'. In the bottom right corner, the text 'Addis Ababa Private Clinics' is displayed.

PRAAMS

Staff Login

Email address

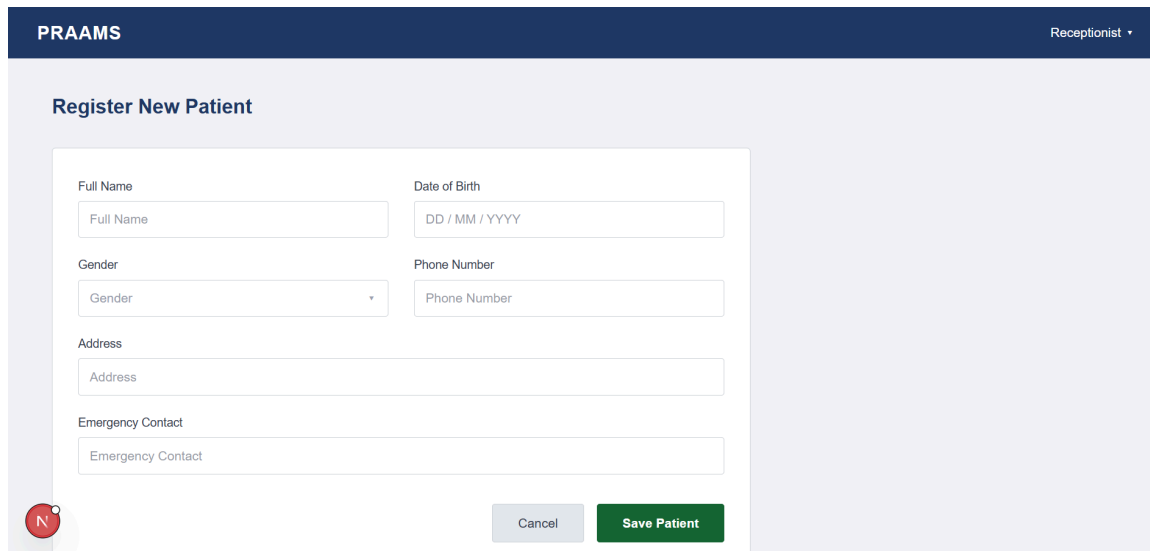
Password

Login

[Forgot password?](#)

 Addis Ababa Private Clinics

Figure 10: Login Screen – Low-Fidelity Mock-up



The mock-up shows a web application for 'PRAAMS' with a dark blue header. The main content area is light gray. A white card in the center is titled 'Register New Patient'. It contains several input fields: 'Full Name' (text), 'Date of Birth' (text with placeholder 'DD / MM / YYYY'), 'Gender' (dropdown menu), 'Phone Number' (text), 'Address' (text), and 'Emergency Contact' (text). At the bottom of the card are two buttons: a gray 'Cancel' button and a green 'Save Patient' button. In the bottom left corner, there is a red circular logo with a white 'N'. In the top right corner of the header, the text 'Receptionist' with a dropdown arrow is displayed.

PRAAMS Receptionist ▾

Register New Patient

Full Name

Date of Birth

Gender

Phone Number

Address

Emergency Contact



Figure 11: Patient Registration Screen – Low-Fidelity Mock-up

PRAAMS
Healthcare Professional

PATIENT	ID	PHONE
Abebe Kebede	P-0231	09xx xxx xxx
Selam Tesfaye	P-0198	09xx xxx xxx

Selam Tesfaye — P-0198, Female, DOB 12/03/1990

MEDICAL HISTORY

14/07/2026 — Follow-up, Dr. Kassa

02/05/2026 — Initial consultation, Dr. Kassa

Figure 12: Patient Search & Record View – Low-Fidelity Mock-up

PRAAMS
Receptionist

Schedule Appointment

Patient

Healthcare Professional

Date

AVAILABLE TIME SLOTS

09:00
09:30
10:00 (booked)
10:30

Reason for visit

Figure 13: Appointment Booking Screen – Low-Fidelity Mock-up

Planned Screen Inventory

Table 19: Planned User Interface Screens

Screen	Primary User(s)	Key Elements
Login Page	All	Email field, password field, login button, error message area

Role-Specific Dashboard	All	Navigation menu, quick stats/summary cards, recent activity
Patient Registration Form	Receptionist	Demographic fields, contact fields, duplicate-check warning, submit button
Patient Search & Record View	Receptionist, Healthcare Professional, Admin	Search bar, results list, patient summary panel, medical history timeline
Medical Record Entry Form	Healthcare Professional	Visit date, diagnosis field, clinical notes field, save button
Appointment Calendar	Receptionist, Healthcare Professional	Day/week/month view, appointment cards, status color-coding
Appointment Booking Form	Receptionist	Patient selector, staff selector, available slot grid, reason field
User Management Panel	Admin	Staff account list, add/edit staff form, role assignment, status toggle
Reports Module	Admin	Report type selector, date range filter, generated report view, export/print controls

Regardless of the screen, the interface will follow the same layout, including a persistent role-aware navigation bar, a primary content region, and form submission success/error messages for form submissions according to the usability requirement in Section 3.3 (NFR-05).

5. References

- HiLCoE School of Computer Science and Technology, *Senior Project Documentation Guideline*.
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6. Annexes

6.1. Data Collection Methods/Tools

The requirements for this system were elicited from semi-structured interviews with clinic staff and a brief structured questionnaire distributed more widely among the clinics, supporting the observations of current (manual) record-keeping and scheduling practices made in section 2.

6.1.1. Interview Questions

Receptionists

1. What is the process involved when a new patient comes in to the clinic.
2. How do you currently look up a file of an existing patient and How long does this typically take?
3. How are appointments scheduled, cancelled and rescheduled?
4. Have you ever had a conflict in the schedule where two patients were due to arrive at the same time, or have conflicting appointments? How was this resolved?
5. What information about a patient would you like to see before they arrive?
6. What is the biggest annoyance or time waster in your day to day record keeping and scheduling duties?
7. If you could change one thing about patient record keeping or appointment scheduling, what would it be?

Healthcare Professionals (Doctors/Nurses)

1. When you see a patient, how do you currently learn their medical history?
2. Have you ever needed to repeat a test or ask a patient to provide additional information because it was not in the record?
3. What types of information do you record after seeing a patient, and how is this information currently recorded?
4. How do you currently keep track of your personal schedule?

5. Are there any concerns you have about patient records being digital rather than written down?

Administrators

1. What reports or statistics do you currently need to review and how are they generated?
2. How are staff permissions managed, if at all?
3. What concerns do you have regarding the security and privacy of patient data?
4. How many staff members and patients are there approximately per month?
5. What would convince you to make the switch to a digital system for these tasks?

6.1.2. Questionnaire Questions

To be distributed to receptionists, healthcare professional and administrators. Questions 1-6 are graded on a scale from 1 to 5 (1 strongly disagree and 5 strongly agree) .

1. Locating a specific patient file is quick and easy with our current system
2. Our appointment system rarely has conflicts or errors
3. I feel confident that patient records are kept secure and private
4. I would feel comfortable using a computerized system to check patients in and keep records, assuming some training was involved
5. Generating reports for administrative purposes is quick and easy with our current system
6. Using a computerized system would help improve the level of care we give to patients
7. On average, how many patients does your clinic register per week? (Open answer)
8. What devices do you think you would use to access a clinic management system? (Desktop computer, Laptop, Tablet, Smartphone, Other)
9. What is your role at the clinic? (Receptionist, Healthcare Professional, Administrator, Other)
10. Is there anything you think a new patient record and appointment system should include? (Open answer)